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Review

Erinacine A and related cyathane diterpenoids: Molecular diversity and mechanisms underlying their neuroprotection and anticancer activities

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ABSTRACT

The presence of a fused 5/6/7 tricyclic core characterizes the group of cyathane diterpene natural products, that include more than 170 compounds, isolated from fungi such as *Cyathus africanus* and *Hericium erinaceus*. These compounds have a common biosynthetic precursor (cyatha-3,12-diene) and can be produced bio- or hemi-synthetically, or via total syntheses. Cyathane diterpenes display a range of pharmacological properties, including anti-inflammatory (possibly through binding to the iNOS protein) and neuroprotective effects. Many cyathanes like cyahookerin C, cyathin Q and cyafranines B and G can stimulate neurite outgrowth in cells, whereas conversely a few molecules (such as scabronine M) inhibit NGF-stimulated neurite outgrowth. The main anticancer cyathanes are erinacine A and cyathins Q and R, with a capacity to trigger cancer cell death dependent on the production of reactive oxygen species (ROS). These compounds, active both in vitro and in vivo, activate different signaling pathways in tumor cells to induce apoptosis (and autophagy) and to upregulate the expression of several proteins implicated in the organization and functioning of the actin cytoskeleton. An analysis of the functional analogy between erinacine A and other natural products known to interfere with the actin network in a ROS-dependent manner (notably cucurbitacin B) further supports the idea that erinacine A functions as a perturbator of the cytoskeleton organization. Collectively, we provide an overview of the molecular diversity of cyathane diterpenes and the main mechanisms of action of the lead compounds, with the objective to encourage further research with these fungal products. The anticancer potential of erinacine A deserves further attention but it will be necessary to better characterize the implicated targets and signaling pathways.

1. Introduction

The cyathane diterpenes refer to a group of natural products possessing an unusual, angularly fused 5/6/7 tricyclic core. They differ from the valparane and mulinane diterpenes which also possess a 5/6/7 tricarbocyclic system (Fig. 1) [1]. The original product, named cyathin, was first isolated from a bird's nest fungus of the genus *Cyathus* (family Nidulariaceae) in the early 1970s [2–5]. The discovery of cyathin or the so called "cyathin complex" from the specie *Cyathus helenae* Brodie collected in the Canadian Rockies [6], opened the way to the discovery of a large family of cyathane diterpenes, including now more than 170 compounds, characterized from numerous *Cyathus* species, such as *C. africanus*, *C. striatus*, and many others [7]. Some of these *Cyathus* species are mainly used in research for the characterization and production of bioactive secondary metabolites, as discussed below; other can be utilized in microbial-assisted pretreatment for the delignification of lignocellulosic substrates, such as *C. stercoreus* used to facilitate carbohydrate accessibility from lignocellulosic biomass [8].

Cyathane diterpenes have been isolated from a variety of fungi, in particular from *Sarcodon* species, such as *S. scabrosus, S. cyrneus* and *S. glaucopus* and from *Phellodon niger* which produces the cyathane-type diterpenoids nigernin A and B. But the main fungus species known to produce a variety of cyathane diterpenes is *Hericium erinaceus* from which have been isolated the erinacines discussed here. The objective of the present review is to highlight the molecular diversity of cyathane diterpenes and their pharmacological properties, with a focus on the anticancer activities and mechanism of action of erinacine A (Fig. 2), one of the lead products in this family of natural products. Cyathanes

Abbreviations: iNOS, inducible nitric oxide synthase; NGF, nerve growth factor; ROS, reactive oxygen species

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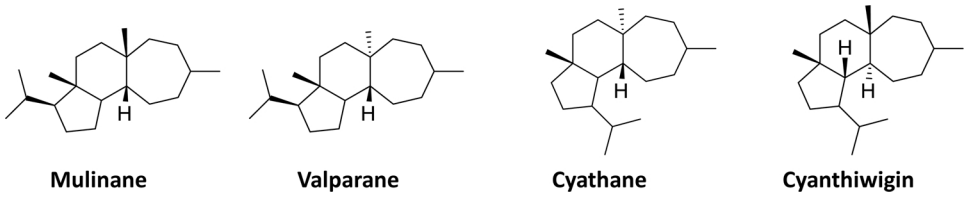

Fig. 1. Structures of the cyathane, valparane and mulinane scaffold found in diterpenes from fungi and the cyanthiwigin scaffold found in marine organisms. They all bear the angularly fused 5/6/7 tricyclic core.

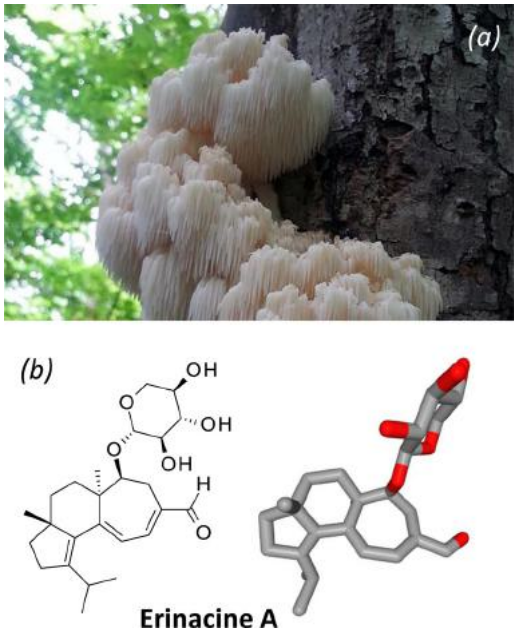

Fig. 2. (a) The fungus *Hericium erinaceus*. Also known as Lion's mane mushroom, hedgehog mushroom (*erinaceus* means hedgehog in Latin), Yamabushitake (Mountain Priest, in Japan), Shishigashira (Lion's head mushroom, in China) or Houtou (Monkey head mushroom, in China). (b) Structure and conformation of erinacine A.

display a range of pharmacological properties such as antibacterial, antimicrobial, fungicidal activities, cytotoxic effects toward cancer cells, anti-inflammatory activities and neuroprotective effects. These aspects are discussed below.

2. Structural diversity of cyathane diterpenes

Over the last 15 years, a variety of cyathane diterpenes have been isolated from different *Cyathus* species. In particular, different bioactive cyathanes were characterized from *C. striatus*, *C. africanus*, and five other *Cyathus* species as indicated in Table 1. New cyathane diterpenes are regularly identified from *Cyathus*, as well as from other species of fungi, notably *Hericium erinaceus* (agaricomycetes), the specie that produces erinacine A. *H. erinaceus* - also known as Lion's Mane mushroom - is considered as a culinary and a medicinal mushroom. It is essentially consumed in Asian countries, for its nutritional and health benefits. This mushroom is a rich source of polysaccharides and secondary metabolites, with more than 70 compounds identified, at the origin of the health-promoting properties of the mushroom fruit bodies and mycelia [9]. Extracts of *H. erinaceus* have revealed diverse pharmacological properties, including antibacterial, anti-inflammatory, anticancer and neuroprotective effects [10]. Many cyathane diterpenes have been isolated from *H. erinaceus* and related species (Table 1). The lead compound is erinacine A (Fig. 1) but there are about 15 naturally occurring erinacine analogues and a small group of cyathane compounds named hericinoids which display modest cytotoxic properties.

Another series of cyathane diterpenes has been isolated also from the bitter mushroom *Sarcodon scabrosus* which produces the scabronines and the related products sarcodonins (Fig. 3). These compounds are interesting due to their anti-inflammatory and neurotrophic properties. Sarcodonin A, neosarcodonin A and derivatives can reduce inflammation induced by phorbol-acetate [11]. The neurotrophic properties of these compounds have stimulated researches to better comprehend their mechanism of action (see below) and therefore important medicinal chemistry studies have been deployed to define the synthetic routes leading to these compounds. For example, several synthetic procedures leading to scabronine G have been reported, as well as the total synthesis of related products such as episcabronine A and scabronine A [12–15]. The total synthesis of sarcodonin G, glaucopine C and cyrneines A–B [16–19], erinacines B and E [20,21], cyathins A–B and derivatives [22,23] have also been described. All these optimized synthetic procedures facilitate the production of multi-gram quantities of cyathane diterpenes necessary for in vivo pharmacology studies or formulation tests. In some cases, the compounds can be produced from scaled-up cultures of the microorganisms. For example, a fermentation process to produce large quantities of erinacine C (> 2.5 g/l) from submerged cultures of *H. erinaceus* has been established [24,25]. Moreover, specific methods have been developed to qualitatively and quantitatively detect cyathane diterpenoids in complex mixtures [24].

We can mention also some rare cyathanes, such as laxitextines A (Fig. 3) and B isolated from the tropical fungus *Laxitextum incrustatum* (found in Kenya). These two compounds display modest antibacterial effects and cytotoxic activities against cancer cells [26]. Two other exotic cytotoxic cyathanes have been isolated from culture of the mycelium of *Strobilurus tenacellus* [27]. In 2015, 105 members were inventoried from various fungi [28] and today our last estimate (04/2020) counted 172 members in this family of cyathane diterpenoids. The family continue to expand. Very recently, erinacine E was found in a new *North American Hericium* specie (strain WBSP8), morphologically similar to *H. americanum* but genetically close to *H. abietis* and *H. alpestra* [29].

The biosynthesis of cyathane diterpenes, from the common isoprene precursor geranylgeranyl pyrophosphate (GGPP) has been well documented, notably the formation of the key intermediate (-)-cyatha-3,12-diene leading to cyatins, sarcodonins, scabronines and erinacines [30,31]. The biosynthesis of erinacines in *H. erinaceum* involves different diterpene cyclases [32] whereas the chemical synthesis of erinacines can be performed using the natural product (+)-carotol as a donor of chirality, via the building block (-)-*trans*-**1**, as illustrated in Fig. 4 [33].

Interestingly, natural products with a cyathane core have been identified also from marine organisms. Indeed, the cyanthiwigin family refers to natural products containing a 5/6/7 tricyclic cyathane unit, first isolated from two species of sea sponge in the early 1990s [34,35]. However, the cyathane diterpenes and the cyanthiwigins can be differentiated by the orientation of the angular methyl groups. The two quaternary methyl groups at C6 and C9 are either disposed *anti* (cyathanes) or *syn* (cyanthiwigins). In addition, the hydrogen atom at C5 relative to the methyl group at C9, is *syn* in the cyathane core and *anti* in the cyanthiwigin core (Fig. 1) [36,37]. Synthetic efforts have been deployed in recent years to facilitate the preparation of the > 20 known cyanthiwigins and the design of analogues [38–42]. Cyanthiwigin F (Fig. 3) displays a limited cytotoxic activity against human primary tumor cells [43] but otherwise there are practically no information

Table 1
Cyathane diterpenoids isolated from fungi.

Fungus species	Cyathane diterpenes isolated	References
Cyathus africanus	Cyathins D-H, R, T, V, W Neosarcodonin O Cyathatriol, 11-O-acetylcyathatriol 3,4- and 4,9-*seco*-cyathanes Neocyathins A-T Cyafricanins A-K	[49,52,53,54,55,101,103,149,150]
Cyathus gansuensis	Cyathins J-P	[151]
Cyathus helenae	Cyathin	[2]
Cyathus hookeri	Cyathin I, erinacine I Cyahookerins A-F	[51,67]
Cyathus stercoreus	Stercorins A-C	[152]
Cyathus striatus	Striatals A-D Striatoids A-F Pyristriatins A-B	[76,96,153,154]
Cyathus subglobisporus	Cyathins A-E 10-hydroxyerinacine S	[155]
Hericium erinaceus	Erinacines A-I,P,Q,T-V,Z1 Hericinoids A-C	[84,85,98,99,156,157,158,159,160,161]
Hericium flagellum	Erinacines A,B,C,E,F, Z1-Z2	[99]
Hericium ramosum	Erinacine E	[162]
Hericium sp. WBSP8	Erinacine E	[29]
Laxitextum incrustatum	Laxitextines A-B	[26]
Phellodon niger	Nigernin A-F	[163,164]
Sarcodon cyrneus	Cyrneines A-E, glaucopine C	[89,90,91,163,164]
Sarcodon glaucopus	Glaucopines A-C	[165–167]
Sarcodon scabrosus	Sarcodonins G,I,L-M Scabronines G,J,M,K,L, Secoscabronine M	[12,80,81,82,83,97,168,169,170]
Strobilurus tenacellus	(12S)- and (12R)-11α,14α-epoxy-13α,14β,15-trihydroxycyath-3-ene	[27]

available about the bioactivity and mechanism of action of these marine cyathanes.

3. Anti-inflammatory activities of cyathane diterpenes

The anti-inflammatory effect of *H. erinaceus*, including erinacine A-enriched extracts, has been well characterized, via a reduction of the production of reactive oxygen species (ROS) and improvement of the antioxidant enzyme catalase and glutathione content [44,45]. But the effects observed with extracts cannot be directly attributed to the cyathane metabolites because they contain also fungal proteins and polysaccharides known to contribute to the anti-inflammatory action [46,47]. Nevertheless, erinacins clearly exhibit anti-inflammatory effects. For example, erinacin A was found to suppress reactive nitrogen species, to reduce the level of pro-inflammatory cytokines such as IL-1β, IL-6 and TNFα and to downregulate the expression of inducible NO synthase (iNOS), p38/MAPK and CHOP in a rat model of ischemia/reperfusion [48]. Similarly, the anti-inflammatory effects of 11-*O*-acetylcyathatriol were attributed to inhibition of iNOS and COX2 proteins, with activation of the MAPK/p38 transduction pathway [49]. Erinacine C exhibits significant anti-inflammatory properties, showing a dose-dependent reduction of pro-inflammatory cytokines such as IL-6, TNFα and NO in a model of inflammation induced by lipopolysaccharide (LPS) in human microglial cells. The compound was found to inhibit of iNOS and NFκB and to activate the Nrf2/HO-1 signaling pathway [50]. Cyathin I also diminishes nitric oxide production in macrophages [51] and the topical application of sarcodonin A inhibits drug-induced inflammation on mouse ears [11].

The anti-inflammatory effects of cyathanes may derive, at least in part, from their capacity to interact with iNOS, the enzyme which produces nitric oxide (NO) from l-arginine. Several cyathanes have been found to modulate the iNOS expression and a direct interaction with the iNOS protein has been proposed. Indeed, a recent molecular docking study has suggested that cyafricanin A binds to the active cavity of iNOs protein [52]. The same proposal was raised before by the same authors studying the cyathanes neocyathins A–J from *C. africanus* [53]. They also showed that another cyathane, allocyathin B_2, inhibited NO production in lipopolysaccharide-stimulated microglia cells [54]. Molecular modeling predicts that the cyathane stercorin B is capable of forming stable complexes with the iNOS protein, implicating two major drug-protein hydrogen bonding interactions, as well as hydrophobic interactions (Fig. 5). There is a drug-binding cavity in the iNOS active site heme pocket known to possibly accommodate different types of naturally occurring diterpenes and steroid glycosides, at least on the basis of molecular modeling studies [55–60]. iNOS induced by inflammatory stimuli and expressed in macrophages is a key mediator of inflammation and host defense systems. The search for potent iNOS inhibitors is a strategy to combat certain bacterial infections, such as *Helicobacter pylori* infections responsible for chronic gastritis and ulcers [61]. Inhibition of iNOS by 11-O-acetylcyathatriol could thus explain the protection from ethanolinduced gastric injury observed with this cyathane derivative [49]. But experimental evidences are awaited to support this proposal only based on molecular modeling. iNOS is a promising target for different Human pathologies (sepsis, heart failure, pain, epilepsy, depression, neurodegenerative disorders) and there is a plethora of potent iNOS inhibitors but no iNOS inhibitors approved for human use [62].

4. Neuroprotection by cyathane diterpenes

Neuroprotection is the main pharmacological characteristic of numerous cyathanes. This property derives from the capacity of certain cyathane diterpenes to induce the release of neurotrophic factors, such as nerve growth factor (NGF) and/or brain-derived neurotrophic factor (BDNF), from glial cells (Fig. 6). For this reason, these compounds are attractive lead molecules for the development of neuroprotective therapeutics to prevent or treat neurodegenerative pathologies, in particular Alzheimer and Parkinson diseases.

The capacity of cyathin A and erinacines to stimulate NGF synthesis and release NGF was recognized early on [63–65]. It was first demonstrated with erinacines A-to-G and later with many other erinacines and related cyathanes [66,67]. One of the most potent cyathanes is

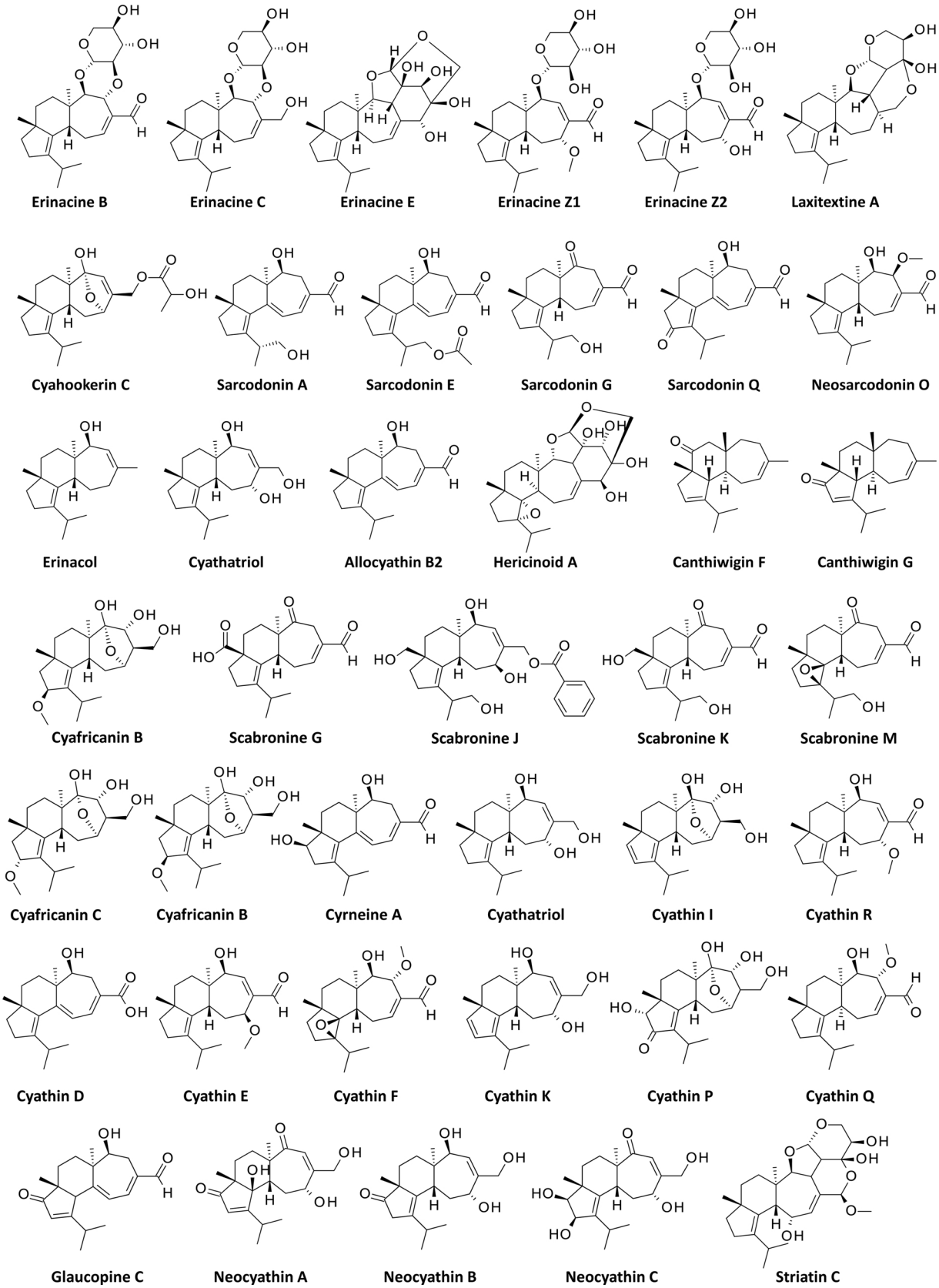

Fig. 3. Molecular structures of different cyathane diterpenes isolated from fungi and the marine products cyanthiwigin F and G.

cyahookerin C (Fig. 3) from *C. hookeri* which was recently found to be the best compound at stimulating neurite outgrowth in NGF-stimulated PC-12 cells within a series of 13 compounds. This natural product enhanced the percentage of neurite-bearing cells in a dose-dependent manner [67]. Other natural products isolated from *H. erinaceus* displays this neurotrophic activity, in particular hericenone derivatives [68–70],

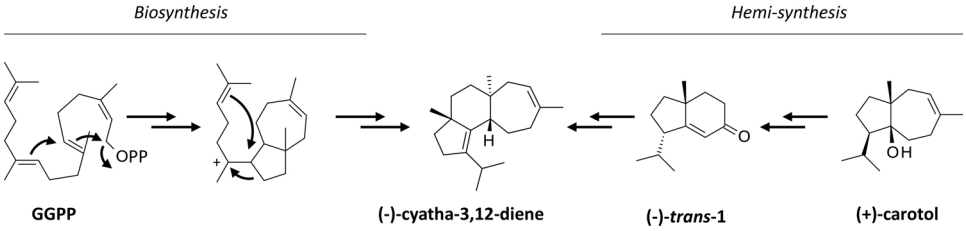

Fig. 4. Structure of (-)-cyatha-3,12-diene as the key intermediate in the biosynthesis of cyathane diterpenes from geranylgeranyl pyrophosphate (GGPP) ([171]], or semi-synthesized from the natural product (+)-carotol, via the intermediate (-)-*trans*-**1** (3-isopropyl-7a-methyl-1,2,3,6,7,7a-hexahydro-inden-5-one) [33].

and the exo-polysaccharide purified from broth of *H. erinaceus* mycelium [71]. For this reason, extracts of *H. erinaceus* are considered very useful to prevent neuronal cell death and to favor axon regeneration [72] as well as to limit the progression of aging and obesity-induced neurodegeneration [73]. *H. erinaceus* is qualified as brain-improving culinary-medicinal mushroom which potently stimulates neurite outgrowth and NGF synthesis [74]. The original product, cyathin A, also induces NGF release from cultured glial cells [75].

Other cyathane diterpenes were found to dose-dependently enhance NGF-mediated neurite outgrowth in cells, such as striatoids A–F [76]. Similarly, scabronines A and G (from *Sarcodon scabrosus*) were found to induce neurite outgrowth of human astrocytoma cells due to the drug-induced increase of the expression of NGF mRNA and concentration-dependent secretion of NGF [77]. The secretion of neurotrophic factors from astrocytoma cells would be due to the activation of protein kinase C (PKC) isoform zeta (PKC-zeta) which then induces the translocation of NFκB to the nucleus, and enhances its transcriptional activity, as shown with scabronine G-methylester [78,79]. But a more recent study indicated that the capacity of the related compound sarcodonin G (as well as sarcodonin A and synthetic derivatives) to influence NGF-induced neurite outgrowth in PC12 cells was both PKC-dependent and -independent and implicated the ERK/CREB signaling pathway as well as phosphorylation of the TrkA receptor [80,81]. Other members of this group of cyathanes from *Sarcodon scabrosus* have been shown to enhance the biosynthesis and secretion of neurotrophic factors, such as scabronine G [12].

The situation is more complex in the scabronine family of cyathanes because opposite effects have been reported. One the one hand, scabronine G-methylester was found to enhance the secretion of neurotrophic factors (NGF and IL-6) from human astrocytoma cells [79]. On the other hand, scabronines K and J showed no significant effect on NGF-stimulated neurite outgrowth in PC12 cells, whereas in the same study sarcodonins A and G were active [82]. Moreover, scabronine M has been found to dose-dependently inhibit NGF-stimulated neurite outgrowth in PC12 cells. The effect was attributed to the drug-induced suppression of phosphorylation of the signaling effectors TrkA and ERK [83]. Therefore, within a homogeneous series of compounds one can find enhancers or inhibitors of neuritogenesis. The same trend has been reported more recently in the sarcodonin series, with synthetic analogues of sarcodonin G capable of inducing or suppressing neurite extension in a primary culture of rat cortical neurons. As illustrated in Fig. 6, the mono-fluorinated derivative 6 promoted NGF-induced neurite outgrowth more potently than sarcodonin G itself, whereas its close structural analogue, the bis-fluorinated derivative 15, suppressed neurite extension in cells. A modeling study suggested that the two compounds do not identically stabilize the NGF-TrkA complex, interacting differently with the NGF ligand or the TrkA receptor, thereby modulating distinctly the downstream signaling ErK/CREB pathway [81].

The case of erinacine A is also interesting because it was found to potentiate NGF-induced neurite outgrowth in PC12 cells but also to induce neuritogenesis in primary rat cortex neurons, possibly via a TrkA-mediated and Erk1/2-dependent pathway [84]. Moreover, a one-month oral treatment with erinacine A attenuated cerebral amyloid-β plaque burden in the brains of 5-month-old APP/PS1 transgenic mice [85]. An erinacin A-enriched *Hericium erinaceus* mycelia (HE-My) or an ethanol extract also ameliorated amyloid-β plaques, increasing the level of insulin degrading enzyme and microglia-mediated clearance [86]. But the mechanism of action of erinacine A seems to be multifactorial and variable from one model to another. In a mouse model of neurotoxicity induced by 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine (MPTP, a model of Parkinson disease), erinacine A treatment was found to reduce neuron damages via activation of a distinct survival pathway, involving the signaling elements PAK1, AKT, LIMK2, MEK, and cofilin [87], as well as activation of the JNK1/2, p38 MAPK and IκB-β/NFκB pathways [88].

Other *Sarcodon* species can enhance neuritogenesis. This is the case of *Sarcodon cyrneus* which produces the cyathane diterpenes called cyrneines A–D, but only cyrneines A–B (not C–D) induce neurite outgrowth [89,90]. A detailed analysis of the neurite outgrowth mechanism triggered by cyrneine A has revealed that the compound activated neither PKC-zeta (unlike the scabronines), nor ERK and JNK signaling. In contrast, the drug was found to activate the GTPase protein Rac1 that regulates actin dynamics. Cyrneine A activated also the transcription factors AP-1 and NFκB and enhanced the expression of the proto-oncogene *c-fos* [91]. Therefore, the induction of neurite outgrowth appears as a property well distributed among cyathane diterpenes but there are apparently different mechanisms leading to the effect, reflecting the complexity of neurogenesis and neural plasticity.

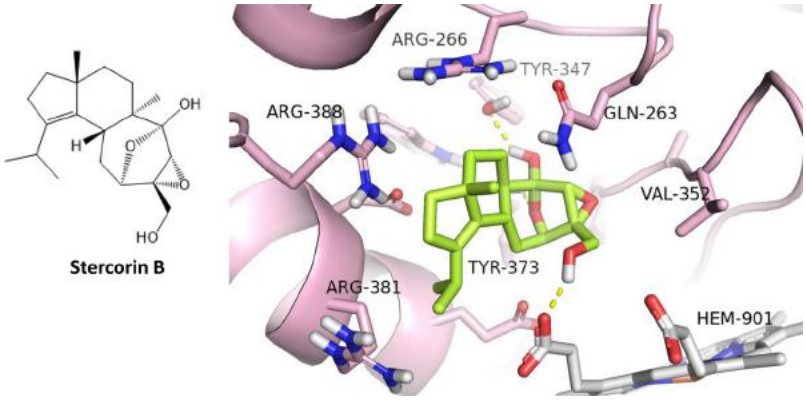

Fig. 5. Molecular model to illustrate the binding of stercorin B to the iNOS protein. The drug fits into a small cavity of the protein delineated by amino-acid residues Arg381, Arrg388, Arg266, Gln263, Val352 and Tyr373. The drug, via its two hydroxyl groups, engages H-bond with Tyr347 and ionic interaction with the heme group. The calculated free binding energy (FBE) predicts a good binding of stercorin B to the iNOS protein (log FBE = -6.58 kcal/mol), with an estimated Inhibition Constant, Ki: 14.98 μM.

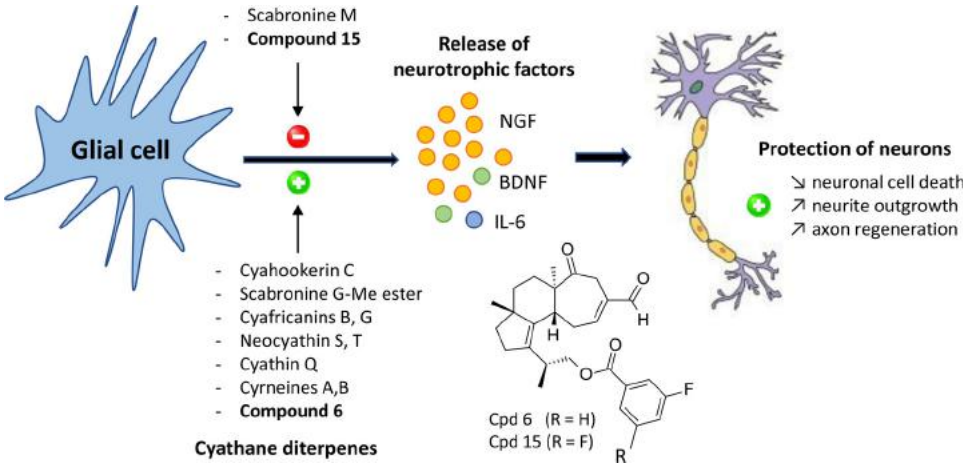

Fig. 6. Different cyathane diterpenes stimulate NGF gene expression and NGF protein transcription, and enhance NGF secretion from glial cells. The secretion of neurotrophic factors (NGF but also BDNF in some cases) protect neurons from cell death and stimulate neurite outgrowth. These effects support the potential use of *H. erinaceus* extracts and potentially certain cyathane diterpenes for the treatment of neurodegenerative diseases such as Alzheimer and Parkinson diseases. In contrast, scabronine M and the synthetic cyathane 15 can inhibit neurite outgrowth in cells [81,83]. The structure of compounds 6 and 15, showing opposite effects on neurite outgrowth, is presented [81].

5. Anticancer properties of cyathane

In 2007, extracts of *C. striatus* were characterized as modulators of the NFκB pathway, with antiproliferative activities against MCF7 breast cancer cells [92]. NFκB is a master transcription factor involved in cancer initiation and progression [93] and a major target for many natural products [94]. Other extracts of *C. striatus* were found to reduce dose-dependently the viability of HPAF-II and PL45 human pancreatic adenocarcinoma cells, with cell cycle arrest and induction of apoptosis [95]. The anti-proliferative activity of the extracts may derive from the various cytotoxic cyathane-type diterpenoids isolated later from this fungus, such as striatals A–D [96] and striatoids A–F [76]. A marked anticancer activity, at least in vitro, has been reported with sarcodonin G, from *Sarcodon scabrosus*. This compound was found to induce apoptosis of HeLa cells, characterized by a marked activation of caspases but it required fairly high concentrations, in the 50–100 μM range, to induce these effects. Sarcodonins E and Q were less active than sarcodonin G [97]. A considerably higher potency was reported in the study of erinacine Z1 and Z2; the later markedly inhibited the growth of HL-60 promyelocytic leukemia cells with a sub-micromolar IC_{50} value (0.5 μM), whereas erinacine Z1 was 18 time less potent (IC_{50} : 8.9 μM) [98]. Z2 is more cytotoxic but Z1 is more efficient than Z2 at promoting the transcription of BDNF and NGF [99]. Erinacines Z1 and Z2 were isolated from *H. erinaceus* and the rare species *H. flagellum* (synonym *H. alpestre*) [99]. Parenthetically, it is worth mentioning here the 4-hydroxy-2-pyridone derivative sambutoxin, isolated from *H. alpestre* which was recently characterized as a potent anticancer agent in vivo, capable of inducing apoptosis via induction of ROS and DNA damages [100]. Although not a cyathane, this case illustrates the large potential of *Hericium* species for drug discovery.

Cytotoxic activities have been reported with other cyathanes, often showing a modest potency with IC_{50} generally in the range of 5–30 μM. For example, neosarcodonin O was found to inhibit the growth of HeLa and K562 cancer cells with IC_{50} of 5.9 and 3.7 μM, respectively, whereas cyathatriol and cyathins D–H were found to be essentially inactive against these tumor cells [101]. The analogue cyathin Q, from *Cyathus africanus*, proved to be more potent and active in vivo. This cyathin derivative inhibited the growth and induced apoptosis of HCT116 and SW5480 colon carcinoma cells and SW480 in vitro, with a much higher efficacy than with three other cancer cell lines (MCF7 breast cancer, HeLa cervical cancer and HepG2 hepatocarcinoma). The mechanism of apoptosis observed with cyathin Q in HCT116 cells was investigated. The drug induced a down-regulation of Bcl-2 and up-regulation of Bim (a master BH3-only Bcl-2 family regulator). Moreover, the use of N-acetyl-cystein as a scavenger of ROS reduced the extent of cyathin Q-induced apoptosis, indicating the implication of ROS in the mechanism of drug-induced cell death (Fig. 7). Interestingly,

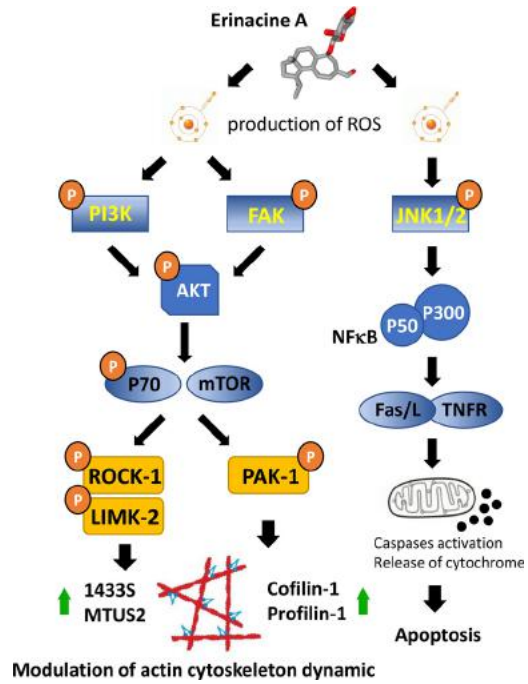

Fig. 7. Schematic of the molecular pathways implicated in the perturbation of the actin cytoskeleton induced by erinacine A [104–107].

the compound also triggered autophagy, characterized by an increased level of the LC3-II marker and an increased expression of the autophagy-related gene ATG-5 (autophagy associated gene 5). Moreover, cyathin Q was able to induce the cleavage of ATG-5 protein which functions as a switch from autophagy to apoptosis and initiation of cell death. It seems that upon drug treatment, the autophagic machinery is activated, probably as a cell survival mechanism, prior to apoptotic cell death finalization in HCT116 cells. The potent cell growth inhibitory and pro-apoptotic effects of cyathin Q in vitro translate into a significant antitumor activity in vivo. A treatment with the drug at 10 mg/kg, intraperitoneally for two weeks led to a marked inhibition of tumor growth in mice bearing HCT116 (and $Bax^{-/-}$ HCT116) tumor xenografts. The tumor volume was reduced by 62 % after two weeks, without any apparent toxicity [102]. This cyathane derivative, which also functions as an inhibitor of NO production and presumed iNOS binder [67], may thus represent a lead compound for the design of anticancer cyathanes.

Important information come also from the mechanistic study of the antitumor activity of cyathin R (from *C. africanus*). This cyathane

derivative was found to induce apoptosis independently of proteins Bax and Bak, by promoting the expression and oligomerization of protein VDAC1 (voltage-dependent anion channel 1), leading to the release of cytotchrome c via the mitochondrial permeability transition pore and apoptosis. Cyathin R exhibits potent anticancer activities in vivo, in HCT116 colon cancer xenograft models [103].

Aside cyathins Q and R, the other potent antitumor cyathane derivative is erinacine A which has been well characterized as a major apoptosis inducer of colorectal cancer cells. This compound target multiple oncogenic pathways, as it is frequently the case with natural products [104]. The signaling cascade activated by the drug implicates the JNK/p300/p50 NFκB pathway, leading to the expression of the TNF receptor as well as Fas and FasL, followed by the activation of caspases and mitochondria activation. A schematic of the drug-activated signaling pathway is depicted in Fig. 7 (adapted from [105]). A potent pro-apoptotic effect was observed in vitro upon treatment of DLD-1 colorectal cancer cells with erinacine A at 30 μM, and in vivo upon treatment of DLD-1 tumor xenografts in mice with erinacine A at 2 − 5 mg/kg/day (administered intraperitoneally). Both in vitro and in vivo, the drug induced a marked activation of gene transcription of the TNF receptor, Fas receptor and Fas ligand [105]. This work revealed the activation of both the intrinsic and extrinsic apoptotic pathways by erinacine A in colorectal cancer cells. The same authors had previously characterized the induction of apoptosis by erinacine A in gastric cancer cells, with a marked activation of the FAK/AKT/PAK1 pathway. Moreover, a proteomic analysis revealed that the drug markedly upregulated the expression of proteins 1433S (14-3-3 sigma protein) and MTUS2 (microtubule-associated tumor suppressor 2) in TSGH9201 gastric cancer cells, presumably via a drug-induced reorganization of the action cytoskeleton which is controlled by the FAK/AKT/PAK1 pathway [106]. The upstream element or signal that triggers all these events is not precisely known but a primary implication of ROS was proposed, based on the capacity of erinacine A to induce the production of ROS and the fact that the inhibition of ROS production (with the scavenger N-acetylcystein) considerably reduced the extent of cell death induced by erinacine A [106]. Increased production of ROS is a common factor for the antitumor cyathanes erinacine A and cyathin Q.

A similar situation has been reported in the course of another proteomic analysis of colorectal cancer cells treated with erinacine A. In this case, the drug upregulated a subset of other proteins, notably cofilin-1 and profilin-1, which are also actin-modulating proteins. Here again, the upregulation of cofilin-1 and profilin-1 was found to be dependent on the drug-induced production of ROS [107]. A mechanism was proposed whereby erinacine A generated ROS which then activated the phospho-PI3K/AKT/mTOR/ROCK1/LIMK2 signaling cascade, leading to the upregulation of cofilin and profilin, and then actin alterations responsible for the inhibition of cancer cells proliferation [107]. Thus, the two proteomic analyses, using different cancer cell types, revealed an activation of distinct signaling pathways but concur to show that erinacine A induces cell death via a perturbation of the actin cytoskeleton dynamic. We strongly suspect that the actin network plays a key role in the antitumor action of different cyathanes (see discussion), just as it is certainly largely implicated in the induction of neurite outgrowth. Indeed, the cyathane cyrneine A was shown to induce actin translocation and accumulation of F-actin at the tip of neurites [91].

6. Other bioactivities

Extracts of *H. erinaceus* display antibacterial properties, notably effects against the gastric pathogen *Helicobacter pylori* [108]. They also show anxiolytic and antidepressant-like effects and prevent neurodegeneration [109]. Erinacine A-enriched extracts of *H. erinaceus* have been elaborated for the treatment of neurodegenerative disorders. This type of mycelia supplement can promote longevity in mice, through the induction of endogenous antioxidants enzymes [110] and improves recognition memory in frail mice [111]. It also displays antidepressant-like effects [112] and stimulates neurite outgrowth [113], while being safe and practically non-toxic for consumption within the appropriate doses [114]. Different clinical trials have been performed to evaluate the safety and improvement of the cognitive functions by taking supplements containing fruiting bodies of *H. erinaceus*. Recently, the oral intake of tablets of *H. erinaceus* was found to be safe, to improve significantly cognitive functions and to prevent from further deterioration [115]. Previous clinical studies had also revealed an improvement in cognitive function upon treatment with *H. erinaceus* tablets [116]. An 8-weeks oral supplementation with *H. erinaceus* tablets decreased depression, anxiety and sleep disorders in subjects affected by overweight or obesity [117]. The clinical effects cannot be linked to the cyathane diterpenes contained in the extracts because the medicinal preparations contain a large variety of bioactive compounds, cyathanes but also aromatic compounds (notably hericenones) [9,118] and polysaccharides also largely exploited for therapeutic and sanitarian applications [119].

7. Discussion

Fungi can be a rich source of medically important compounds, such as penicillins, cyclosporin A and lovastatin, to cite only a few. They remain a largely untapped chemical resource for drug discovery [120,121]. The fungal biodiversity is huge, offering a vast potential for the identification and design of novel medicinal products. In this context, fungal cyathane terpenoids represent an underexplored family of natural products, not extensively studied and probably underestimated in terms of therapeutic potential. In the Asian culture, there is a long history for the traditional medical use of lion's mane mushroom (*Hericium erinaceus*) to treat symptoms of Qi deficiency (such as insomnia, vacuity and loss of strength) as well as to promote good digestion, to sustain brain and nerve health, to reinforce vigor and strength, to fortify the spleen and to nourish the gut [122]. Diverse natural products contained in lion's mane mushroom extracts and preparations are at the origin of these activities, such as polysaccharides, sterols, and terpenoids including many cyathane diterpenes. At mentioned above, clinical trials with *H. erinaceus* extracts have now been performed, showing that the product is safe and effective to attenuate depression and anxiety syndromes [117]. It is not known if the cyathanes contained in such extracts contribute to the anxiolytic and antidepressant-like effects observed in Human with the extracts [123] but it is clear that cyathanes can exert potent neuroprotective effects, notably via their capacity to enhance NGF-stimulated neurite outgrowth and to inhibit the production of pro-inflammatory cytokines, via inhibition of proteins like iNOS and possibly COX-2. In fact, extracts of *H. erinaceus* display multiple bioactivities, including anti-microbial, anti-inflammatory, antioxidant, immune-modulatory effects and others [10] but the contribution of cyathane diterpenes is essentially centered around the neuroprotective and anticancer properties.

The neuroprotection activity of cyathane diterpenes derives from their capacity to modulate the stimulatory effect of NGF, via an interference with the TrkA-NGF receptor-ligand complex. It is known for a long time that cyathanes represent useful amplifiers of the neurite stimulatory activity of NGF, as is the case with other natural products, such as 6-shogaol, andrographolides and many other compounds [124,125]. But it is also clear now that cyathanes can also inhibit the stimulatory activity of NGF, as underlined with scabronine M and cyahookerin derivatives [67,83]. Cyathane diterpenes thus represent tools to modulate, positively or negatively, the NGF/TrkA interaction, with a direct consequence on neuritogenesis (Fig. 6). It is tempting to propose a link between the capacity of erinacine A to promote NGF-induced neurite outgrowth [84] and the improvement of the depression symptoms observed with *H. erinaceus* extracts enriched with erinacine A [126]. Along these lines, the Chinese herbal decoction Kai-Xin-San, used in traditional medicine to treat depression for thousands of years,

has been found to potentiate NGF-induced neurite outgrowth through an upregulation of the expression and activity of TrkA [127,128]. But in fact, the mechanism of action of erinacine A is obviously multifactorial, implicating not only modulation of the NGF and BDNF pathways [111], but also interference with the mitochondrial functioning [129], leading to an increased production of reactive oxygen species and cell cytotoxicity [107]. Apparently, there is a balance of pro- vs. anti-oxidative activities of erinacine-containing products, because erinacine A-enriched *H. erinaceus* mycelia strongly induce anti-oxidative enzyme activities of superoxide dismutase, catalase and glutathione peroxidase, contributing to the decrease of toxin-induced neuronal cell cytotoxicity and apoptosis [87,88,109]. In contrast, erinacine A itself can increase cytotoxicity and the generation of reactive oxygen species in cancer cells [106,107]. But the anticancer activity of erinacine A cannot be simply summarized in terms of ROS production, iNOS binding (which remains a hypothesis based on modeling studies) and induction of apoptosis. There is a lack of knowledge in terms of potential molecular targets. A thorough understanding of the molecular targets of erinacine A (and cyathane diterpene in general) and the mechanisms of action underlying its anti-cancer effects is necessary to promote its use as a new scaffold to design anticancer drugs. For this reason, at present erinacine A does not necessarily show a superior advantage compared to other anticancer natural products like curcumin, thymoquinone or honokiol [130–132] but it represents a distinct lead for the design of more potent analogues.

In this context, the recent discovery of a potential action of erinacine A on microtubule dynamic and the cytoskeleton reorganization is very interesting and informative. Indeed, the proteomic analysis revealed that erinacine A upregulates the expression of MTUS2 (Microtubule Associated Scaffold Protein 2) which is also known as TIP150, a microtubule plus-end-tracking protein. TIP150 interacts with and targets the mitotic centromere-associated kinesin (MCAK) at the plus-ends and regulates kinetochore microtubule plus-end dynamics in cell division [133,134]. Erinacine A was found also to upregulate the sigma isoform of 14-3-3 protein (14-3-3σ, stratifin) which has oncogenic properties [135]. Looking for other drugs capable of upregulating 14-3-3σ, we identified the antitumor triterpenoid cucurbitacin B (Fig. 8) known to activate expression of 14-3-3σ in a ROS-dependent manner in cancer cells [136]. Moreover, this tetracyclic triterpenoid is known (i) to interact with cytoskeletal proteins such as microtubules and actin filaments [137], (ii) to induce intracellular ROS generation and activation of the FAK pathway [138] and (iii) to react with cellular thiols to mediate its anti-cancer effects [139]. Although the two drugs erinacine A and cucurbitacin B are structurally distinct (Fig. 8), they seem to behave similarly, via a disruption of the actin cytoskeleton and ROS/FAK activation. Unlike cucurbitacin B, erinacine A (and cyathin Q) is not known to react with thiols but however, it is likely a reactive molecule through its α,β-unsaturated formyl moiety. Other fungal metabolite with a similar functional group are known to form covalent adducts with proteins or glutathione (e.g. ophiobolin A) [140]. The parallel between erinacine A and the cucurbitacins is significant. An enhanced expression of actin binding proteins such as cofilin-1 was observed with both erinacine A and cucurbitacin B [141]. Several studies have established that cucurbitacins interfere with actin dynamics in a ROS-dependent manner, and with an implication of the mTOR pathway [142,143], similarly to what has been proposed with erinacine A. Both drugs were also found to reduce the expression of nucleophosmin.

A few other natural products have been shown to upregulate protein 14-3-3σ, notably the microtubule-stabilizing drugs Taxol and epothilone B which are well-established antitumor agents [144]. Another interesting natural product to mention here is the cyclic monoterpene α-phellandrene (Fig. 8) which was found to increase ROS production and 14-3-3σ expression in cancer cells [145,146]. Coincidentally or not, this natural product bears a structural resemblance with the isopropylcyclopentene side of erinacine A which may thus represent the actin-interacting side of the natural product. Finally, we would like also to mention the potential analogy between erinacine A and the sesterpenoid fungal phytotoxin ophiobolin A (calmodulin inhibitor). This anticancer agent not only bears an identical reactive α,β-unsaturated formyl moiety (reacting with cysteinyl thiols) but also produces endogenous ROS and alters the dynamic organization of the F-actin

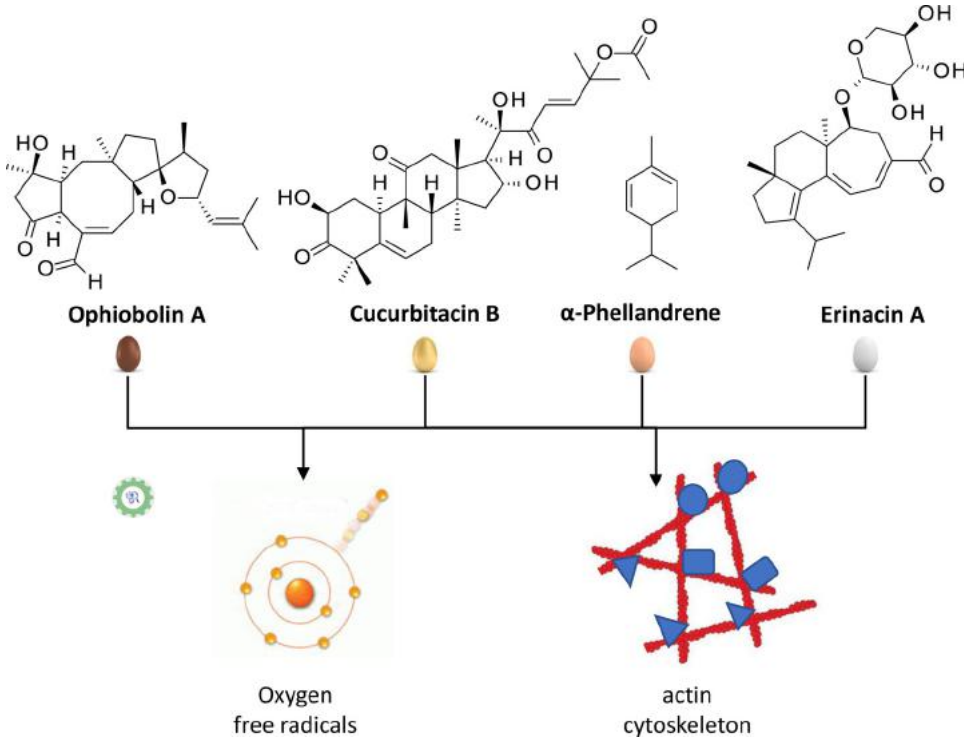

Fig. 8. Functional analogy between 4 different natural products known to enhance ROS production in cells and to perturb the actin cytoskeleton organization and dynamic: erinacine A, cucurbitacin B, α-phellandrene and ophiobolin A.

cytoskeleton [147,148] (Fig. 8). Collectively, these different observations led us to raise the hypothesis that anticancer cyathanes like erinacine A and cyathin Q function as perturbators of the actin cytoskeleton organization, via a ROS-dependent microtubule-associated mechanism.

In conclusion, cyathane diterpenes represent a large family of natural products essentially isolated from fungi. They function as regulators of neurite outgrowth and a few compounds, such as erinacine A and cyathins R and Q, also display interesting anticancer activities. The molecular targets of these compounds are not known at present, but their ROS-dependent activities lead to a marked perturbation of key components of the actin cytoskeleton and induction of apoptotic cell death.

Declaration of Competing Interest

The authors declare no conflicts of interest associated with this publication and there has been no significant financial support for this work that could have influenced its outcome.

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